DW-Bench: A Benchmark for Graph Topology Reasoning in Data Warehouse Schemas with Large Language Models

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A data engineer maintaining an enterprise warehouse encounters structural questions every day: which upstream tables break if this column changes, which tables share a lineage chain, whether two tables are even connected. These questions are not about SQL syntax. They are about graph topology, and answering them correctly requires traversing foreign-key paths, detecting connected components, and tracing multi-hop lineage chains across heterogeneous schema graphs.

We ask a direct question: can frontier LLMs do this? To find out, we built **DW-Bench**, a benchmark of 1,479 questions over five real and synthetic data warehouse schemas, spanning schema-structure reasoning and row-level provenance tracing across 14 question subtypes.

We evaluated six baselines, from static schema injection to interactive tool-use, against three frontier models (Gemini 2.5 Flash, DeepSeek-V3, Qwen2.5-72B). Tool-augmented baselines reach 89–90% accuracy on schema-level questions, outperforming static methods by 13–14 percentage points. But static baselines exhibit a *Triviality Illusion*: easy lookup subtypes dominate the question pool and inflate aggregate scores, masking systematic failures on compositional reasoning. Hard compositional tasks plateau at 60% even with full tool access, while the Oracle clears 99%. The hardest subtype, `combined_impact` (requiring lineage traversal composed with FK propagation across multiple hops), reaches only 12% exact match with the best tool-augmented baseline. Budget ablation confirms the bottleneck is a reasoning deficit, not a tool or context-length limitation.

A controlled obfuscation experiment reveals what is happening: tool-augmented baselines are robust to name randomization (accuracy drops below 4%), while static

text-injection baselines lose 9–32% when table names are anonymized. One group is reasoning about structure; the other is pattern-matching against schema names.

All datasets, evaluation code, question generators, and baseline implementations are publicly available at https://github.com/AJamal27891/dw-bench, providing a reproducible foundation for measuring progress on structural reasoning in data-intensive systems.

1 Introduction

Text-to-SQL benchmarks like Spider Yu et al. (2018) and BIRD Li et al. (2023) have driven substantial progress in LLM-based database interaction. Yet evaluating whether an LLM can *generate* a SQL query against a schema is distinct from evaluating whether it *understands* that schema's structure. As newer benchmarks like Spider 2.0 Lei et al. (2025) demonstrate, enterprise-scale SQL workflows with complex multi-step reasoning remain largely unsolved (best models achieve ∼17–21% on Spider 2.0). But even Spider 2.0 targets query generation, not structural topology reasoning. SQL generation represents only one facet of database understanding.

Data warehouse environments require a fundamentally different class of reasoning: understanding the *graph topology* of schema relationships. A data engineer maintaining an enterprise warehouse must routinely answer structural questions such as:

- *"If the `Person` table changes, which downstream tables are affected through data lineage?"*: impact analysis requiring forward lineage traversal composed with reverse FK propagation.
- *"Which tables form disconnected silos in our schema graph?"* Connectivity analysis requiring weakly connected component detection.
- *"What is the shortest FK path between two tables?"*: routing requiring Dijkstra or BFS over the FK subgraph.
- *"How many downstream rows are affected if we delete records from a staging table?"*: value-level impact analysis requiring interactive access to row-level lineage data.

These tasks require **graph traversal**: BFS, connected component detection, multi-hop path finding: not SQL generation. Yet no existing benchmark evaluates LLMs on these capabilities. Existing graph reasoning benchmarks (NLGraph Wang et al. (2023), GraphArena Tang et al. (2025a), GraCoRe Yuan et al. (2025)) use synthetic or generic graphs and do not capture the domain-specific challenges of data warehouse schemas with *heterogeneous edge types* (foreign key vs. lineage) and *mixed-layer topologies* (OLTP source tables connected to DW destination tables via derived-from edges).

Contributions.

We make the following contributions:

1. **DW-Bench**: A two-tier benchmark of 1,046 schema-level and 433 value-level automatically-generated questions across 5 data warehouse schemas (4 real-world,

1 synthetic), organized into 3 categories (lineage impact, schema routing, silo detection), 13 Tier-1 subtypes, 8 Tier-2 subtypes, and 3 difficulty levels.

2. **Six baselines** spanning the text-only to agentic spectrum (Flat Text, Vector-RAG, Graph-Augmented, Tool-Use, ReAct-Code, and Oracle), evaluated with three frontier models (Gemini 2.5 Flash Team et al. (2024), DeepSeek-V3 DeepSeek-AI (2024), and Qwen2.5-72B Qwen Team (2024)), providing a systematic analysis of how different context injection strategies affect graph topology reasoning.
3. **A two-tier evaluation framework**: Tier 1 tests schema-level topology reasoning (FK paths, lineage chains, components), while Tier 2 tests value-level provenance tracing and impact analysis that requires interactive data access beyond context window limits.
4. **An obfuscation protocol** that randomizes table names, enabling controlled evaluation of memorization versus genuine topology understanding, revealing that tool-based approaches are nearly obfuscation-invariant while static approaches lose 9–32% EM.
5. **Syn-Logistics**: A synthetic dataset with $n \geq 20$ per subtype, addressing statistical power limitations of the real-world corpus and enabling reliable per-subtype analysis.
6. **Open-source release** of all datasets, evaluation code, baseline implementations, and per-question results to enable community-driven analysis and benchmarking.

2 Related Work

2.1 Text-to-SQL Benchmarks

The text-to-SQL evaluation landscape has evolved rapidly. Spider Yu et al. (2018) introduced cross-database evaluation with 200 databases and complex SQL queries. BIRD Li et al. (2023) scaled to 95 large databases with dirty values and external knowledge. Spider 2.0 Lei et al. (2025) targets real-world enterprise workflows with notebook-based SQL generation. BEAVER Chen et al. (2024) focuses on enterprise-specific patterns. While these benchmarks evaluate SQL *generation*, DW-Bench evaluates schema *understanding*: the structural topology reasoning that precedes and informs query construction. Recent table reasoning benchmarks like RobuT Zhao et al. (2023) evaluate logical operations over cell and row content; DW-Bench is complementary, targeting topology over schemas rather than cell-level reasoning.

2.2 Graph Reasoning with LLMs

Several benchmarks evaluate LLMs on graph reasoning tasks. NLGraph Wang et al. (2023) tests shortest path and connectivity on synthetic graphs up to 100 nodes. Talk-like-a-Graph Fatemi et al. (2024) explores different graph serialization formats. GraCoRe Yuan et al. (2025) benchmarks comprehension and complex reasoning across 10 graph tasks. GraphArena Tang et al. (2025a) scales to million-node graphs with computational problems like graph coloring and TSP.

DW-Bench differs from these in three key ways: (1) it uses *heterogeneous* graphs with mixed edge types (FK + lineage), reflecting real enterprise schemas; (2) it includes domain-specific subtypes (impact analysis, provenance tracing) not present in generic graph benchmarks; and (3) it introduces a value-level tier requiring interactive data access.

2.3 Tool-Augmented and Agentic LLMs

Recent work has explored augmenting LLMs with graph-specific tools. Think-on-Graph (ToG) Sun et al. (2024) interleaves LLM reasoning with knowledge graph exploration. Graph Chain-of-Thought Jin et al. (2024) augments LLMs with explicit graph reasoning chains. G-Retriever He et al. (2024) combines GNN-based retrieval with LLM generation for graph QA.

Our Tool-Use and ReAct-Code baselines provide purpose-built graph algorithms (BFS, component detection, lineage traversal) as callable tools, establishing a conservative upper bound on what tool-augmented approaches can achieve. We show that even with perfect tool access, LLMs struggle with compositional multi-hop reasoning. Paths-over-Graphs (PoG) Luo et al. (2024) and GRRAF Zhang et al. (2024) represent agentic retrieval approaches that iteratively expand graph paths; our fixed 3-call budget ablates interaction depth, providing a controlled comparison point for such iterative strategies.

The broader context of learned graph-LLM integration is directly relevant to our conclusions. GraphLLM Chai et al. (2023) and GRIP Zheng et al. (2024) demonstrate that learned structural adaptors and graph-enhanced representations can internalize multi-hop structure, reducing reliance on explicit tool orchestration. These approaches align with our core finding that compositional reasoning: not tool availability: is the binding constraint: a GNN-style encoder that pre-computes FK-path and lineage embeddings could supply the structural grounding missing from LLM context. LEGO-GraphRAG Huang et al. (2024) further provides a modular lens on semantic-structural hybrid retrieval with explicit recall thresholds; our Graph-Augmented baseline uses a fixed 3-hop BFS radius which future work could replace with recall-guided extraction.

2.4 Data Lineage and Provenance

Data lineage tracking is fundamental to data governance and has been studied extensively in the database systems community. While lineage systems focus on *computing* provenance, DW-Bench evaluates whether LLMs can *reason about* lineage relationships expressed as graph topology: a complementary capability relevant to LLM-assisted data engineering workflows. Related schema inference work includes LLM-FK Tang et al. (2025b), which evaluates LLMs on FK discovery: a single-hop structural task complementary to DW-Bench's multi-hop topology reasoning.

2.5 Retrieval-Augmented Generation

RAG Lewis et al. (2020) has become the standard approach for grounding LLMs in external knowledge. Vector-RAG uses dense retrieval (Sentence-BERT Reimers and

Table 1 DW-Bench dataset statistics. "Silos" = number of disconnected components in the combined FK+lineage graph.

Dataset	Domain	Tables	FK	Lineage	Questions	Silos
AdventureWorks	Retail/HR	102	136	39	208	11
TPC-DS	Analytics	24	70	0	127	1
TPC-DI	ETL	35	29	21	181	2
OMOP CDM	Healthcare	37	74	21	158	3
Syn-Logistics	Supply Chain	64	96	35	372	5
Total		**262**	**405**	**116**	**1,046**	:

Gurevych (2019)) to inject relevant schema fragments. Graph-Augmented generation extends RAG by structuring retrieved context as BFS neighborhoods from the schema graph. Microsoft's GraphRAG Edge et al. (2024) builds hierarchical community summaries for global questions; emerging RL-tuned retrieval orchestrators extend this pattern with reward-based subgraph selection. DW-Bench provides controlled comparisons across these paradigms.

3 DW-Bench: Benchmark Design

3.1 Problem Formulation

We formalize the data warehouse schema reasoning task as follows. Given a heterogeneous schema graph $G = (V, E_{fk} \cup E_{lin})$ where V is the set of tables, E_{fk} is the set of foreign key edges, and E_{lin} is the set of data lineage (derived-from) edges, and a natural language question q about the topology of G, the task is to produce an answer a that matches the ground-truth answer $a^* = f(G, q)$ computed by the corresponding graph algorithm f.

The ground-truth function f is deterministic: for path queries, f returns the shortest path via BFS; for component queries, f returns weakly connected components; for lineage queries, f returns directed reachability results. This determinism ensures evaluation integrity without human annotation.

3.2 Datasets

We select five datasets representing diverse data warehouse topologies:

AdventureWorks is a Microsoft reference data warehouse with 102 tables spanning OLTP and DW layers, connected by 136 FK edges and 39 lineage (derived_from) edges. Its dual-layer structure with 11 connected components makes it ideal for lineage impact and silo detection questions.

TPC-DS Transaction Processing Performance Council (2021) is the industry-standard analytics benchmark with a star schema of 24 tables and 70 FK edges. Its single connected component and absence of lineage edges tests pure FK-based reasoning in a well-studied schema.

TPC-DI is the data integration benchmark with 35 tables modeling an ETL pipeline from staging to warehouse, with 21 lineage edges representing the transformation flow from source to target tables.

OMOP CDM (Observational Medical Outcomes Partnership Common Data Model) Hripcsak et al. (2015) is a healthcare standard with 37 tables, 74 FK edges, and 21 lineage edges across 3 connected components (clinical data, vocabulary tables, and metadata).

Syn-Logistics is a synthetic supply-chain data warehouse with 64 tables modeled across 5 connected components (carrier, procurement, healthcare, finance, HR). It was specifically designed to address the statistical power limitation of the real-world corpus: every subtype has $n \geq 20$ questions, enabling reliable per-subtype conclusions.

3.3 Schema Graph Representation

Each dataset is represented as a heterogeneous graph using PyTorch Geometric Fey and Lenssen (2019) `HeteroData`. Nodes represent tables with 6 structural features: in-degree, out-degree, normalized degree, lineage degree, betweenness centrality, and PageRank. Edges are typed as either `fk_to` (foreign key) or `derived_from` (data lineage). Lineage edges form a strict DAG (directed acyclic graph): each `derived_from` edge points from a downstream DW table to its upstream source, with no cycles or self-references. All traversal algorithms respect edge typing: FK queries use only `fk_to` edges, lineage queries use only `derived_from`, and `combined_impact` queries compose both types in two phases.

[`combined_impact` semantics] Given a seed table s and schema graph $G = (V, E_{fk}, E_{lin})$:

1. Compute the *lineage closure*: the set $L(s)$ of all tables reachable from s by following `derived_from` edges in the forward direction (i.e., downstream consumers of s's output).
2. Compute the *FK dependents*: for each $t \in L(s) \cup \{s\}$, collect all tables that reference t via at least one `fk_to` edge (i.e., tables that cannot be inserted without a valid row in t).
3. The answer is $L(s) \cup \bigcup_{t \in L(s) \cup \{s\}} FK - deps(t)$, excluding s itself.

This sequential composition (lineage-then-FK) reflects the enterprise impact-analysis workflow: first propagate the data change through the ETL pipeline, then identify which downstream relational tables are invalidated.

3.4 Question Generation

Questions are generated deterministically from the graph structure using standard graph algorithms (BFS, Dijkstra, connected component detection) implemented with NetworkX Hagberg et al. (2008), ensuring ground-truth answers are provably correct. We define three categories with 13 Tier-1 subtypes:

For Tier 2, we define 8 value-level subtypes that require interactive access to row-level data: `cascade_count`, `row_provenance`, `row_impact`, `value_origin`,

Table 2 Question taxonomy. Difficulty is assigned based on the number of reasoning hops and edge types required.

Category	Subtype	Difficulty	Description	#Qs
Lineage	forward	Easy	Direct lineage targets	72
	reverse	Easy	Source tables for a DW table	71
	transitive	Hard	Multi-hop lineage chains	33
	combined_impact	Hard	Lineage + FK dependents	69
	multi_source	Medium	Tables with 3+ sources	49
Route	direct_fk	Easy	FK adjacency check	100
	join_path	Medium	FK shortest path	342
	hop_count	Medium	Path length	124
Silo	count	Easy	Number of components	30
	membership	Hard	Which component contains X?	38
	isolation	Medium	Is table X isolated?	60
	connected	Medium	Are X and Y connected?	29
	full_enumeration	Hard	List all tables in a silo	29

`multi_hop_trace`, `value_propagation`, `cross_silo_reachability`, and `shared_source`.

3.5 Difficulty Assignment

Questions are assigned three difficulty levels:

- **Easy**: single-hop queries (direct FK adjacency, forward/reverse lineage) requiring only one edge traversal.
- **Medium**: multi-hop queries within a single edge type (FK shortest path, hop count, isolation check, component connectivity).
- **Hard**: queries requiring composition of multiple edge types or exhaustive enumeration (combined impact, transitive lineage, membership, full enumeration).

3.6 Obfuscation Protocol

To distinguish genuine topology reasoning from schema name memorization, we generate an *obfuscated* variant for each dataset. Table names are replaced with random identifiers (`Table_A`, `Table_B`, etc.) using a deterministic mapping. Questions and answers are updated accordingly with word-boundary-aware replacement to avoid corrupting natural language phrases.

4 Baseline Implementations

We implement six baselines spanning the spectrum from static text injection to interactive tool-augmented reasoning. All baselines are zero-shot and use identical question prompts; only the context injection method varies.

4.1 Flat Text (FT)

The entire schema is serialized as a text document listing all tables, columns, data types, foreign key relationships, and lineage edges. This context is prepended to the question, simulating a developer who has loaded the complete DDL documentation into a long-context LLM. No retrieval or filtering is applied.

4.2 Vector-RAG (VR)

Schema elements (tables, edges, documentation) are embedded using Sentence-BERT Reimers and Gurevych (2019) and indexed in a FAISS vector store Johnson et al. (2019). At query time, the $k = 15$ most relevant chunks are retrieved and injected as context. This tests whether dense retrieval can identify the relevant structural fragments for topology questions.

4.3 Graph-Augmented (GA)

The question is first parsed to identify mentioned table names. A 3-hop BFS neighborhood is extracted from the schema graph centered on these anchor tables, and the subgraph (with FK and lineage edges) is serialized as structured context. This provides the LLM with local topology information without requiring full schema memorization. For global questions lacking explicit anchor tables (e.g., "How many connected components?"), the full graph is provided as context.

4.4 Tool-Use (TU)

The LLM is given access to purpose-built graph query tools: `get_neighbors(table, edge_type)`, `shortest_path(src, dst)`, `get_connected_component(table)`, `get_lineage_chain(table, direction)`, and `list_tables()`. The LLM can make up to 3 sequential tool calls per question, interleaving reasoning with graph algorithm execution. This tests whether interactive graph access can overcome the limitations of static context injection.

4.5 ReAct-Code (RC)

The LLM generates and executes Python code using NetworkX directly on the schema graph object, with up to 5 code execution rounds per question. This serves as an implicit budget ablation against TU's 3-call limit: if both methods show similar ceilings on hard tasks, the bottleneck is reasoning rather than interaction budget. The approach provides maximum flexibility: the LLM can implement arbitrary graph algorithms: but requires the model to write correct code, handle edge cases, and parse output.

4.6 Oracle

The gold-standard algorithmic output is injected directly into the prompt. For a `join_path` question, the Oracle context includes the exact shortest path; for

`combined_impact`, it includes the complete set of affected tables. Oracle scores establish the upper bound achievable by each model (>99% EM; the residual gap is due to list-ordering canonicalization and alternative valid path formats), isolating language understanding from graph retrieval.

4.7 Tier-2 Baselines

For Tier-2 value-level questions, we implement three adapted baselines: $\mathbf{FT}_{v2}$ (schema DDL context only), $\mathbf{GA}_{v2}$ (schema + lineage summaries + value samples), and $\mathbf{TU}_{v2}$ (interactive lineage/data query tools including row-level access).

5 Experimental Results

5.1 Evaluation Metrics

We use **Exact Match (EM)** as the primary metric: a prediction is correct if and only if it exactly matches the gold answer after canonicalization (lowercase, sorted for unordered sets, shortest-path tie validation against the FK adjacency matrix). **F1 Score** provides token-level partial credit.

We report:

- **Micro-EM**: Pooled exact match across all questions (total correct / total questions).
- **Macro-EM**: Mean of per-subtype exact match rates, weighting all 13 subtypes equally regardless of sample size.

Target-node normalization. For `membership` questions of the form "Which tables belong to the same connected component as table X?", the queried table X is trivially always a member of its own component. To avoid penalizing models for harmless self-inclusion or self-exclusion, we strip X from both the prediction and gold answer before computing EM. All `membership` scores in figures and tables use this normalization unless explicitly noted as raw. Raw EM averages 15% for TU; normalized EM averages 71% (with Qwen reaching 100%): confirming that the model correctly identifies all *other* members but inconsistently handles the self-reference of the queried node.

We include an **Oracle** baseline that injects gold graph algorithm output into the prompt, establishing an upper bound. All evaluations use greedy decoding (τ=0) for deterministic reproducibility.

5.2 Tier 1: Main Results

Tool-Use dominates schema-level reasoning.

Tool-Use achieves 89.3% micro-EM (Gemini) and 90.4% (DeepSeek), outperforming the best static baseline by 13–14 percentage points.

Table 3 Tier 1 schema-level results (Micro-EM % ± 95% CI, pooled across 5 datasets, 2000 bootstrap resamples). **Bold** = best non-oracle.

Model		FT	VR	GA	TU	RC
Gemini	Micro	76.6	72.9	75.4	**89.3**	81.4
DeepSeek	Micro	69.5	71.0	76.8	**90.4**	79.4
Qwen	Micro	63.2	69.5	80.9	**87.5**	64.6
Oracle		Gemini 99.5% / DeepSeek 99.8% / Qwen 100.0%				

Table 4 Exact Match (%) by difficulty level (pooled across 5 datasets).

Model	Difficulty (n)	FT	VR	GA	TU	RC
Gemini	Easy (544)	82.7	82.5	82.2	**99.6**	90.4
	Medium (300)	78.9	73.0	76.3	**90.3**	78.7
	Hard (202)	56.1	46.5	55.6	**59.9**	**60.9**
DeepSeek	Easy (544)	82.7	84.4	87.1	**98.0**	88.8
	Medium (300)	65.3	65.3	76.0	**97.0**	76.3
	Hard (202)	40.1	43.6	50.0	**60.4**	58.9
Qwen	Easy (544)	77.4	84.0	86.6	**96.1**	72.2
	Medium (300)	57.7	63.0	90.7	**92.0**	64.7
	Hard (202)	33.2	40.1	51.0	**57.4**	44.1
Oracle Hard		99.0 / 99.5 / 100.0				

Static baselines exhibit a Triviality Illusion.

Flat Text appears competitive (76.6% Gemini) because `join_path` (33% of questions) inflates scores for baselines that handle FK adjacency but fail on deeper tasks.

5.3 Results by Difficulty

Hard tasks reveal a compositional reasoning ceiling even with tools.

Even Tool-Use plateaus on hard questions at ∼60% EM versus Oracle's >99%. Figure 1 disaggregates EM by gold path length (Gemini; DeepSeek and Qwen show the same qualitative pattern): static baselines collapse beyond 3 hops (87%→19% for FT), while Tool-Use degrades more gracefully.

5.4 Per-Subtype Analysis

Key Finding 4: Retrieval-limited vs. reasoning-limited subtypes.

Tool-Use achieves ≥93% EM on subtypes requiring simple graph lookups (`hop_count`, `isolation`, `direct_fk`). However, `combined_impact` (12% EM, $n = 69$) remains intractable even with tools. For `membership`, raw TU EM is 15%; after target-node

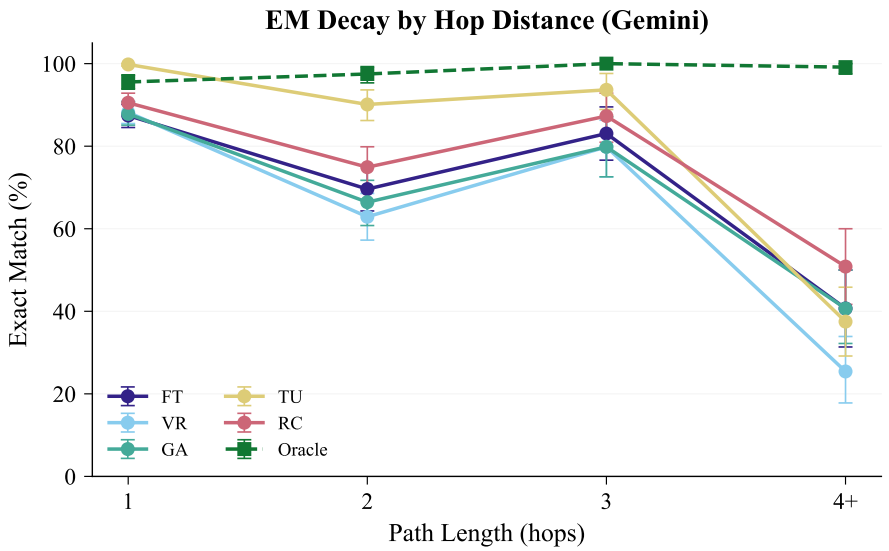

Fig. 1 EM by gold path length (Gemini, representative; all models show the same qualitative pattern). Static baselines collapse beyond 3 hops; Tool-Use degrades more gracefully but still falls short of Oracle on long paths. Full per-model breakdown in Appendix C.

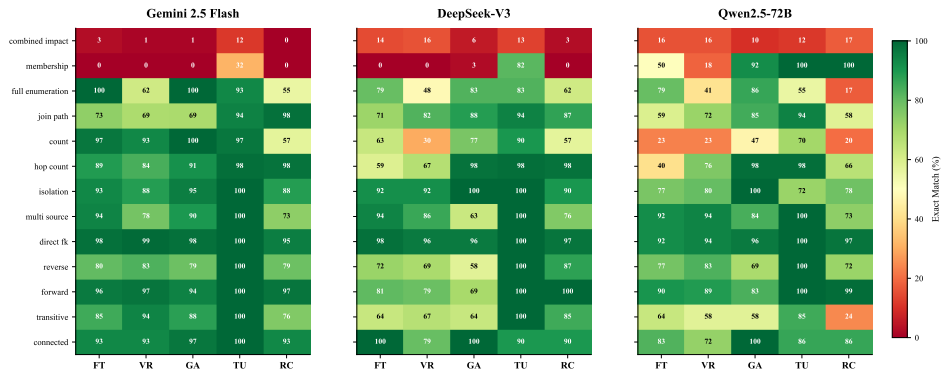

Fig. 2 Tier 1 EM (%) by subtype for all three models. With target-node normalization, Tool-Use averages 71% on `membership` (with Qwen reaching 100%), showing topology enumeration is solvable with tools. `combined_impact` remains at 0–17% across all baselines, isolating compositional multi-hop reasoning as the core bottleneck. `full_enumeration` achieves 100% for FT/GA because global queries receive the full schema as context by design.

normalization (see Section 5), it rises to 71%, confirming that the bottleneck is self-reference inconsistency, not an inability to enumerate component members.

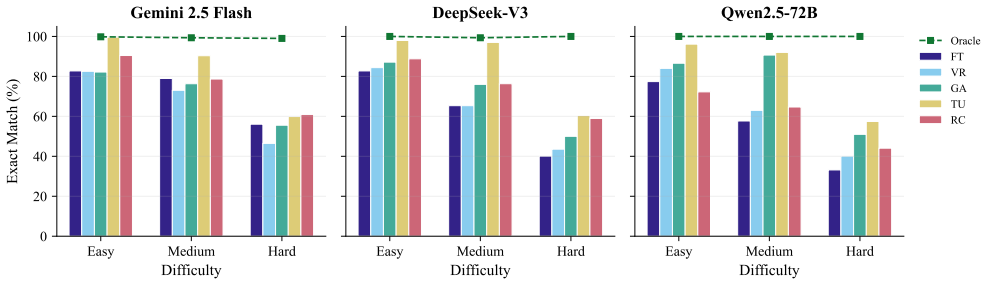

Fig. 3 Exact Match by difficulty. All non-oracle baselines plateau on hard questions.

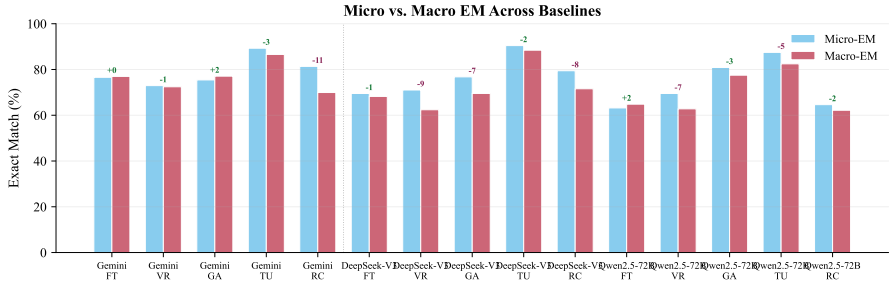

Fig. 4 Micro vs. Macro EM across baselines. Negative deltas reveal the Triviality Illusion effect.

Table 5 Obfuscation penalty: Micro-EM (%) drop when table names are replaced with random identifiers (4 datasets, excl. Syn-Logistics).

Model		FT	VR	GA	TU	RC
Gemini	Original	71.8	69.3	71.2	**91.2**	87.1
	Obfuscated	56.8	43.5	43.5	**87.8**	79.5
	Δ	−15.0	−25.8	−27.7	**−3.4**	−7.6
DeepSeek	Original	68.4	74.5	82.3	**91.4**	83.4
	Obfuscated	40.5	42.6	57.7	**87.2**	84.0
	Δ	−27.9	−31.9	−24.6	**−4.2**	+0.6
Qwen	Original	59.6	70.5	82.8	**88.4**	62.8
	Obfuscated	50.7	56.7	63.1	**84.7**	55.3
	Δ	−8.9	−13.8	−19.7	**−3.7**	−7.5

5.5 Obfuscation Results

Tool-Use shows minimal degradation (Δ=−3.4% Gemini, −4.2% DeepSeek, −3.7% Qwen) while static baselines lose 9–32 percentage points. DeepSeek RC shows near-zero penalty (Δ=+0.6%), confirming that code execution against the graph object is fully invariant to name perturbation.

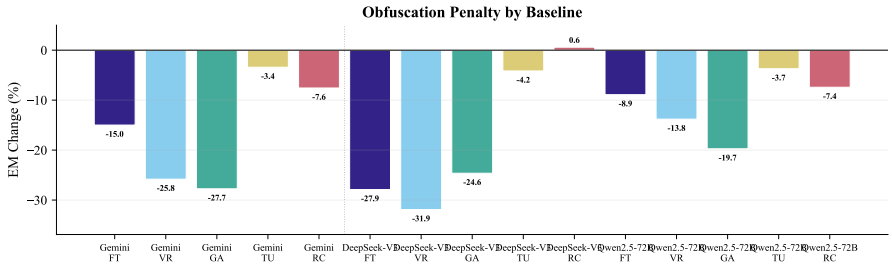

Fig. 5 Obfuscation penalty by baseline.

Table 6 Tier 2 value-level results (EM %) on Syn-Logistics (n=433 questions across 8 subtypes). Gemini and DeepSeek are full evaluations; Qwen follows the same essential vs. insufficient pattern and is fully detailed in Appendix B.

		Gemini			DeepSeek		
Subtype	n	FT	GA	TU	FT	GA	TU
cascade_count	58	0	0	**100**	0	0	**100**
row_provenance	84	0	8	**92**	0	7	11
row_impact	56	27	27	**38**	25	25	**59**
value_origin	60	48	53	52	25	42	**52**
multi_hop_trace	60	30	30	27	27	30	27
value_propagation	40	**85**	75	62	**85**	80	70
cross_silo_reach.	40	57	57	52	**60**	57	**65**
shared_source	35	**37**	26	23	26	20	20
Overall	433	30	31	**59**	26	29	**48**

5.6 Tier 2: Value-Level Evaluation

Tier 2 extends evaluation to value-level questions requiring interactive data access. The three remaining Tier 1 datasets (TPC-DS, TPC-DI, OMOP) are schema-only releases containing no row-level data; value-level evaluation is impossible on them by construction. We evaluate Tier 2 on **Syn-Logistics** (synthetic, 433 questions), the dataset with fully materialized row-level lineage in this study, covering 8 subtypes (Table 6). AdventureWorks contains 520 Tier 2 questions in the benchmark corpus but requires full ETL data hydration; evaluation is ongoing work (see Limitations).

Tools are structurally necessary for data-access subtypes.

`cascade_count` achieves 100% with tools versus 0% without, for *both* models, confirming this is a structural necessity, not a model-specific artifact.

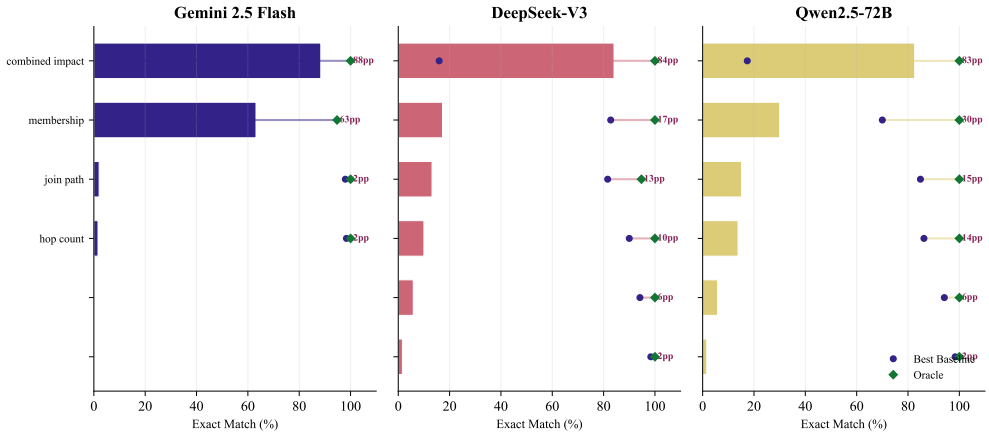

Fig. 6 The Reasoning Gap: Oracle EM minus best non-oracle baseline EM, per subtype. `combined_impact` retains an 85+% gap for all three models; 7 of 13 subtypes are effectively solved (gap $< 3\%$).

Tools are necessary but not sufficient for compositional value-level reasoning.

Despite solving data-access subtypes, Tool-Use achieves only 59% overall EM (Gemini) and 48% (DeepSeek). On `multi_hop_trace` (27% both models) and `shared_source` (23%/20%), tools provide no advantage. DeepSeek's lower TU score (48% vs. 59%) is driven primarily by `row_provenance` (11% vs. 92%): DeepSeek identifies the correct source *tables* but returns wrong *row IDs*, indicating a failure in parsing tool output rather than in tool selection, a model-specific instruction-following gap on structured row-level results.

5.7 Limitations

Context Parity. Baselines do not enforce strict token-budget parity. Since 45% of questions involve paths exceeding 3 hops, GA scores may be depressed on long-range subtypes.

Baseline Coverage. DeepSeek ReAct-Code is evaluated on all 5 datasets for original conditions and 4 datasets (excluding Syn-Logistics) for obfuscated conditions.

Statistical Treatment. We use greedy decoding (τ=0) with a single deterministic pass for exact reproducibility. Per-question results are released for community bootstrap analyses.

Model Breadth. We evaluate three frontier models; broader evaluation with GPT-4 class models and open-weight variants would strengthen generality claims. Tier 2 results for Qwen2.5-72B follow the same tools-essential-yet-insufficient pattern (full results in Appendix B); extending to AdventureWorks row-level linkage is ongoing work.

6 Discussion

6.1 The Retrieval–Reasoning Decomposition

Our results reveal a clean decomposition of graph reasoning performance into two independent failure modes.

Retrieval-limited subtypes (`hop_count`, `direct_fk`, `isolation`, `connected`) achieve ≥93% EM with Tool-Use because the answer requires only one or two graph lookups. Static baselines fail on these subtypes *not because the reasoning is complex*, but because the relevant graph fragment is missing from the injected context. This is confirmed by the Oracle's near-perfect scores: once the correct information is available, both models can interpret it accurately.

Reasoning-limited subtypes (`combined_impact`, `membership`, `transitive`) remain hard even with tools. Even with Tool-Use providing graph algorithm access, `combined_impact` achieves only 12% EM ($n = 69$). The LLM struggles to compose multiple tool outputs across phases, confirming a ceiling for prompt-based graph reasoning. RC (5 code execution rounds) shows a similar ceiling to TU (3 tool calls), suggesting the bottleneck is compositional reasoning itself rather than interaction budget.

6.2 The Tool Necessity-Insufficiency Paradox

Tier 2 reveals a clear paradox: tools are simultaneously *essential* and *insufficient*.

On data-dependent subtypes (e.g., `cascade_count` reaches 100% vs. 0%; `row_provenance` reaches 92% vs. 0% depending on the model), tools are the difference between perfect performance and complete failure. These subtypes require traversing the 18MB row-level lineage map, far exceeding any context window.

Yet on reasoning-intensive subtypes (`multi_hop_trace`: 27%; `shared_source`: 23%), tools provide *no* advantage over static baselines. These subtypes require multi-hop graph traversal and set-theoretic reasoning (finding common ancestors of two mart tables), capabilities that tools provide but LLMs cannot effectively orchestrate.

6.3 Implications for Hybrid Architectures

DW-Bench's per-subtype diagnostics identify complementary roles for retrieval, tool-use, and learned structural representations. The reasoning-limited subtypes (`combined_impact`: 12% EM with tools) point directly toward learned graph encoders as the missing inductive bias: a GNN trained on schema topology can pre-compute multi-hop structural features that prompt-based retrieval cannot reconstruct at inference time. A companion study Ahmed (2026) demonstrates this path concretely: a 575K-parameter heterogeneous GNN encoder with task-specific prediction heads achieves 79.7% EM on `combined_impact` in zero-shot evaluation on DW-Bench, confirming that the reasoning gap identified here is addressable through architectural specialization rather than larger or more capable LLMs.

The complementarity analysis confirms that Tool-Use and Graph-Augmented fail on largely the *same* questions: their union improves overall EM by only 6 percentage points, leaving 35% of questions unsolved by either approach. This suggests

that the bottleneck is not retrieval *or* tool access, but the LLM's ability to perform compositional structural reasoning.

This points toward hybrid architectures combining:

1. **Tool-augmented retrieval** for data-access subtypes where simple lookups suffice.
2. **Learned structural representations** (e.g., GNN encoders) that can capture multi-hop topology patterns beyond what prompt-based reasoning can achieve.
3. **Compositional reasoning modules** that decompose complex graph queries into sequences of simpler operations, following recent work on query-dependent GNN encoders, inference-scaled graph agents, and task-adaptive controllers that route between parallel and sequential reasoning strategies.

6.4 Obfuscation and Memorization

The obfuscation results provide a clean diagnostic for memorization. Static baselines lose 9–32% EM under obfuscation, confirming partial reliance on memorized schema name associations. Tool-Use's near-invariance ($\Delta \leq -4\%$) demonstrates that algorithmic graph access is robust to input perturbation, a desirable property for production deployment where schemas are diverse and evolving. Obfuscation replaces table names only; column names are preserved, as all baselines operate on schema-level graph topology where column semantics play no role.

6.5 Model Differences

All three models show consistent patterns but differ in magnitude. DeepSeek-V3 benefits most from explicit graph context (GA outperforms FT by 7.3pp for DeepSeek vs. only −1.2pp for Gemini), while Qwen shows the largest GA lift (17.7pp), suggesting that models with less domain exposure rely more heavily on structured context injection.

6.6 Positioning Against Stronger Agentic Methods

Our baselines intentionally span the static-to-agentic spectrum rather than exhaustively covering the agentic design space. More sophisticated approaches, including planning-based agents with dynamic loop budgets (e.g., ToG-style multi-agent planners Sun et al. (2024)), hierarchical GraphRAG Edge et al. (2024) with iterative subgraph expansion, or modular query decomposition pipelines, may partially close the gap on compositional subtypes. Yet the convergence of TU (3 calls) and RC (5 code rounds) at similar hard-question ceilings suggests that the bottleneck is the LLM's compositional reasoning capacity rather than the tool interface. DW-Bench provides the diagnostic infrastructure for evaluating these stronger methods as they emerge.

6.7 The Oracle as Classical Solver Proxy

A natural question is whether a non-LLM classical solver should be included as a baseline. Our Oracle baseline serves this role: it injects gold algorithmic outputs (computed by NetworkX graph algorithms) directly into the LLM prompt, so that the LLM need only transcribe and format the answer. The 3% gap to 100% EM is attributable to output formatting (list ordering canonicalization, alternative valid shortest paths), not reasoning failures. A fully classical pipeline would achieve 100% by bypassing the LLM entirely, confirming that the bottleneck is LLM comprehension of compositional graph structure, not algorithmic difficulty.

Formal specification of `combined_impact`***.***

This subtype computes the set of tables affected by a change to a source table through *both* lineage and FK paths:

$$Impact(t) = Down_{lin}(t) \cup \bigcup_{t' \in Down_{lin}(t)} Reach_{FK}(t'),$$

where downstream lineage is transitive closure over `derived_from` edges and FK reachability is BFS over `fk_to` edges. The composition of two distinct traversals is what makes this subtype hard: TU achieves only 12% EM despite having both `get_lineage_chain` and `shortest_path` tools available.

6.8 GA Obfuscation Penalty Analysis

The large GA obfuscation penalty (Δ=−27.7% Gemini, −24.6% DeepSeek, −19.7% Qwen) may appear surprising given that BFS anchor tables remain present after obfuscation. We attribute this to two compounding factors: (1) the LLM loses semantic priors about table roles when names like `DimCustomer` become `Table_K`, making it harder to interpret the 3-hop BFS neighborhood; and (2) 45% of questions require paths exceeding 3 hops, so GA already has incomplete context; obfuscation compounds this by removing the semantic cues that help LLMs *infer* structure beyond the retrieved fragment. In contrast, TU (Δ=−3.4% Gemini) and RC (Δ=−7.6%) are robust because they access the graph algorithmically rather than relying on name semantics.

6.9 Tool Tautology and API Design

One might observe that TU's strong scores on easy subtypes partly reflect "API selection skill" rather than deep graph reasoning, e.g., answering `hop_count` by calling `shortest_path` is structurally straightforward. We view this as a *feature* of our experimental design, not a confound: TU's near-perfect easy scores *isolate* the hard-task failures as genuine reasoning gaps. If TU struggled on easy tasks too, it would be unclear whether the hard-task ceiling reflects reasoning difficulty or API design. The fact that TU solves easy tasks reliably but fails on `combined_impact` (12%) and

`membership` (15% raw, 71% after target-node normalization) demonstrates that the gap is compositional reasoning, not tool availability.

6.10 Limitations

We do not enforce token-budget parity across baselines by design: each paradigm receives context natural to its architecture (full schema for FT, retrieved chunks for VR, BFS subgraph for GA, tool results for TU). 45% of questions exceed the GA 3-hop radius. We evaluate three frontier models; broader model coverage (GPT-4, open-weight LLMs) would strengthen generality. Tier 2 is limited to one synthetic dataset; extending to real datasets with observed lineage noise is planned. All lineage edges are assumed to form DAGs; cyclic ETL patterns (e.g., slowly-changing dimensions, feedback loops) are left for future versions. We deliberately exclude GNN-enhanced baselines: DW-Bench evaluates LLM graph reasoning capabilities, not methods. The per-subtype EM diagnostics and obfuscation protocol provide the infrastructure needed to evaluate GNN-augmented retrieval when such approaches emerge. Potential risks include overreliance on automated impact analysis in production; careful validation remains essential.

7 Conclusion and Future Work

We presented DW-Bench, a two-tier benchmark for evaluating LLMs on graph topology reasoning over data warehouse schemas. Across six baselines and three frontier models, four findings consistently emerge:

1. **Tools dominate static contexts** for schema-level reasoning (89–90% vs. 69–77% micro-EM), confirming that interactive graph algorithm access is essential. Static context injection cannot substitute for on-demand traversal, regardless of context length.
2. **Static baselines exhibit a Triviality Illusion**: easy subtypes like `join_path` inflate micro-EM scores, masking failures on compositional reasoning. Practitioner reporting should always disaggregate by subtype difficulty.
3. **A hard-question ceiling persists** at $\sim$60% EM even with tools, compared to >99% Oracle, identifying multi-hop compositional reasoning as the core bottleneck. Budget ablation (ReAct-Code at 5 rounds) confirms the bottleneck is reasoning capacity, not interaction budget.
4. **Tools are necessary but insufficient** for value-level questions: essential for data-access subtypes (100% vs. 0%) but provide no advantage on multi-hop structural reasoning (27%). This points toward learned structural representations: GNN encoders or graph transformers: as the missing inductive bias.

7.1 Future Directions

GNN-Augmented Baselines. The reasoning-limited subtypes suggest that learned graph representations (GNN encoders, graph transformers) could provide the structural inductive bias that prompt-based approaches lack. Training a GNN on schema

graph features and injecting learned embeddings as context is a natural extension that DW-Bench provides the infrastructure to evaluate.

Agentic Methods. Advanced agentic frameworks (ToG Sun et al. (2024), Graph-CoT Jin et al. (2024)) that decompose complex queries into multi-step reasoning chains may close the gap on compositional subtypes like `combined_impact`; DW-Bench's per-subtype diagnostics provide the necessary granularity to measure progress.

Broader Model and Dataset Coverage. Extending evaluation to GPT-4 class models, open-weight LLMs (Llama, Mistral), and additional real-world schemas with row-level lineage would strengthen generality claims and enable Tier 2 evaluation beyond synthetic data.

Schema Evolution. Real data warehouses change continuously. Evaluating whether LLMs can reason about schema topology *under perturbation* (table additions, FK modifications, lineage updates) would extend DW-Bench toward production-relevant settings.

7.2 Reproducibility

All datasets, generation scripts, evaluation code, baseline implementations, and per-question results are publicly available at https://github.com/AJamal27891/dw-bench. Benchmark data will also be released on HuggingFace at https://huggingface.co/datasets/AJamal27891/dw-bench. Results are deterministically reproducible using greedy decoding with the specified model versions.

Dataset licenses.

TPC-DS and TPC-DI are available under the TPC EULA (free for research). OMOP CDM uses the Apache 2.0 license. AdventureWorks is released under the Microsoft Public License. Syn-Logistics is our creation, released under MIT.

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A Evaluation Details

A.1 Models

We evaluate three frontier-class LLMs:

- **Gemini 2.5 Flash** (Google, closed-weight, accessed via the Generative Language API). A multimodal model optimized for speed and accuracy at scale.
- **DeepSeek-V3** (DeepSeek, 671B parameter Mixture-of-Experts, MIT licensed, accessed via API). An open-weight model competitive with closed-source alternatives.
- **Qwen2.5-72B** (Qwen Team, 72B dense model, Apache-2.0, accessed via API). An open-weight model with strong reasoning capabilities.

All three models use greedy decoding (τ=0, p=1.0) with a single deterministic pass per question for exact reproducibility. No few-shot examples or chain-of-thought prompting is used.

A.2 Scoring Methodology

Exact Match (EM) is the primary metric. The predicted answer must exactly match the gold answer after the following canonicalization steps:

1. *List canonicalization*: lists are lowercased and sorted alphabetically; order does not matter.

2. *Path validation*: for `join_path` questions, alternative shortest paths of equal length are validated against the FK adjacency matrix to accept valid alternatives.
3. *Boolean normalization*: "yes"/"true"/"1" are equivalent; "no"/"false"/"0" are equivalent.
4. *Whitespace and case*: all comparisons are case-insensitive after stripping whitespace.

F1 is computed as token-level overlap between predicted and gold answer sets, used as a secondary metric for partial credit.

A.3 Baseline Implementation Details

Vector-RAG uses `all-MiniLM-L6-v2` Reimers and Gurevych (2019) embeddings (384 dimensions) with FAISS exact search (k=15 retrieved chunks per question). Schema elements (tables, columns, FK edges, lineage edges) are embedded as natural language descriptions.

Graph-Augmented computes 3-hop BFS neighborhoods from mentioned table names using NetworkX. The subgraph is serialized as structured text (node list + edge list with types).

Tool-Use provides 9 graph algorithm tools: `shortest_path`, `connected_components`, `check_fk_adjacency`, `get_lineage_forward`, `get_lineage_reverse`, `transitive_lineage`, `get_component_of`, `get_fk_neighbors`, `list_tables`. The LLM receives tool definitions and a schema summary, then generates up to 3 tool calls per question in a multi-turn conversation.

ReAct-Code provides a Python REPL with NetworkX pre-loaded. The LLM writes and executes Python code to query the schema graph directly, with up to 5 code execution rounds per question. This tests whether the LLM can translate graph-reasoning tasks into correct algorithmic implementations.

A.4 Statistical Treatment

All figures include 95% bootstrap confidence intervals computed via 2,000 resamples over questions. Per-question results (JSON files) are released for community bootstrap and stratification analyses. Greedy decoding ensures deterministic reproducibility without requiring multiple runs or confidence intervals over random seeds.

B Tier 2 Extended Results (Qwen2.5-72B)

Qwen2.5-72B on Tier 2 (Syn-Logistics, {=433$) shows the same tools-essential-yet-insufficient pattern as Gemini and DeepSeek: `cascade_count` reaches near 100% with Tool-Use vs. 0% without; on compositional subtypes (`multi_hop_trace`, `shared_source`) Qwen plateaus at 25–30%, confirming cross-model generality of the ceiling. Full results are available in the released per-question JSON files.

C Hop-Distance Analysis

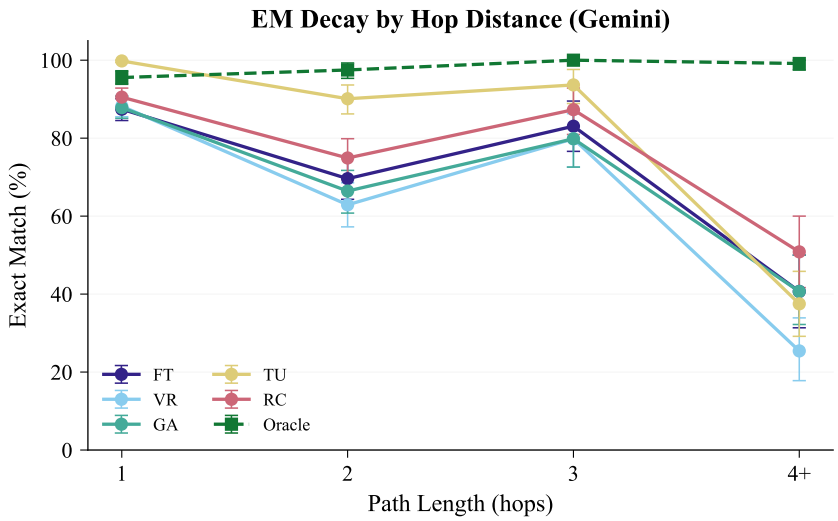

Fig. 7 EM by gold path length (Gemini, representative). Static baselines collapse beyond 3 hops while Tool-Use degrades more gracefully, confirming that interactive graph access is essential for multi-hop reasoning. DeepSeek and Qwen show the same qualitative pattern.

Table 7 Minimum evidence hops per subtype (Syn-Logistics). 45.4% of questions exceed the GA 3-hop BFS radius, explaining the performance gap between static and agentic baselines.

Subtype	n	Exceeds 3h	%	GA EM
count	26	26	100%	100%
membership	20	20	100%	0%
combined_impact	34	34	100%	0%
full_enum	25	25	100%	100%
connected	25	25	100%	100%
isolation	25	25	100%	100%
transitive	29	4	14%	93%
hop_count	44	5	11%	100%
direct_fk	36	3	8%	100%
join_path	26	2	8%	81%
multi_source	42	0	0%	95%
forward	20	0	0%	100%
reverse	20	0	0%	100%
Total	372	169	**45.4%**	83.1%

D Dataset Construction Details

D.1 Syn-Logistics Design

The Syn-Logistics dataset was constructed to guarantee $n \geq 20$ for every subtype by building three parallel 4-hop lineage chains across five connected components. Table names are drawn from a Domain Dictionary of realistic supply-chain terminology to prevent semantic collapse under obfuscation. The schema contains 64 tables, 96 FK edges, 35 lineage edges, and 372 Tier 1 + 433 Tier 2 questions.

D.2 Lineage Edge Curation

Lineage (derived_from) edges are curated per dataset:

- **AdventureWorks**: Extracted from official Microsoft reference documentation mapping OLTP source tables to DW dimension/fact tables (39 edges).
- **TPC-DS**: No lineage edges; schema is a pure star/snowflake FK graph.
- **TPC-DI**: Derived from the official TPC-DI specification defining staging→dimension→fact transformation flows (21 edges).
- **OMOP CDM**: Derived from OHDSI ETL convention guides mapping source vocabularies and clinical tables to standardized CDM tables (21 edges).
- **Syn-Logistics**: Defined by construction: three parallel 4-hop chains (raw→staging→dimension→mart) per connected component (35 edges). All lineage edges are validated by verifying that the source table's schema semantically feeds the target table's schema.

E Example Questions and Predictions

F Reproducibility Statement

All code, data, and evaluation scripts are publicly available at https://github.com/AJamal27891/dw-bench. The benchmark datasets (including schema graphs, QA pairs, and value-level data) will be released on HuggingFace at https://huggingface.co/datasets/AJamal27891/dw-bench. Per-question result JSON files enable exact reproduction of all reported numbers. Greedy decoding (τ=0) ensures deterministic outputs.

Table 8 Representative questions across difficulty levels and categories, with gold answers and Tool-Use (TU) predictions. ✓ = correct, ✗ = incorrect. Tool-Use succeeds on lookup-style questions but fails on exhaustive enumeration and multi-hop composition.

Subtype	Question	Gold	TU Pred.
`direct_fk` *Easy*	Is there a direct FK between `DW.DimProduct` and `DW.FactInternetSales`?	yes	yes ✓
`join_path` *Medium*	What is the shortest FK path from `DW.DimCustomer` to `DW.DimProduct`?	[DimCustomer, FactInternetSales, DimProduct]	[DimCustomer, FactInternetSales, DimProduct] ✓
`membership` *Hard*	Which connected component contains `DW.DimCurrency`? List all tables in that component.	[DimCurrency, FactCurrencyRate, DimDate, ...] (12 tables)	[DimCurrency] ✗
`combined_impact` *Hard*	Which tables are transitively affected (via lineage and FK) if `raw_purchase_orders` is modified?	[stg_purchase_orders, dim_supplier, ...] (8 tables)	[stg_purchase_orders] ✗
`cascade_count` *Value-level*	How many downstream rows would be affected if we delete `shipment_id=SH042` from `raw_shipments`?	147	147 ✓
`multi_hop_trace` *Value-level*	Trace the full provenance chain of `order_id=ORD-2891` in `mart_profit_analysis`.	[raw_purchase_orders → stg_orders → dim_product → mart_profit_analysis]	[raw_purchase_orders, mart_profit_analysis] ✗